

Solution Requirements Specification: Functional, Non-Functional & Service Level Agreements

Project: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Document Reference: REQ-PHASE2-SRS-V1.0

Target Environment: ServiceNow Utah / Vancouver / Washington DC / Xanadu LTS

System Module: Service Catalog, Flow Designer, ITAM, Platform Governance

1. Executive Summary

This document establishes the definitive engineering baseline for the automated Standard Laptop Procurement solution. It details:

- 1. **Functional Requirements (FR-01 to FR-12):** The deterministic business capabilities, routing logic, validation policies, and transactional workflows required.
- 2. **Non-Functional Requirements (NFR-01 to NFR-06):** The performance, security, availability, usability, auditability, and maintainability mandates.
- 3. **Platform Constraints & Architecture Assumptions:** The operational boundaries governing ServiceNow configuration.
- 4. **SLA & OLA Governance Framework:** The multi-tiered Service Level Agreement and Operational Level Agreement hierarchy ensuring end-to-end fulfillment within 72 business hours.

2. Functional Requirements (FR)

The system enforces 12 discrete functional requirements across the requisition, governance, staging, asset management, and lifecycle closure modules.

Requirement ID	Requirement Name	Description & Functional Logic	Priority	Acceptance Verification Criteria
FR-01	Unified Catalog Item Definition	The system shall provide a single catalog item named "Standard Laptop Order" located under Hardware > Computers in the enterprise Service Catalog. The item shall be governed by User Criteria (snc_internal), accessible to all active corporate employees.	Must Have	Authenticate as standard employee; verify item appears in catalog search and category view; verify external/vendor users cannot access item.

FR-02	Structured Variable Architecture & Dynamic Defaults	The catalog form shall capture: <code>requested_for</code> (Reference: <code>sys_user</code>), <code>hardware_bundle</code> (Choice: Windows Business, Windows Developer, macOS Developer), <code>delivery_method</code> (Choice: Desk Drop, Depot Pickup, Remote Courier), <code>shipping_address</code> (String), and <code>business_justification</code> (Multi-line text). Form shall dynamically auto-populate employee department, site, and manager from <code>sys_user</code> .	Must Have	Load form as employee Alex Chen; verify <code>requested_for</code> Department, Location, and Manager auto-fill; verify <code>shipping_address</code> become visible and mandatory only when <code>delivery_method</code> = <code>'Remote Courier'</code> .
FR-03	Transactional Order Cart & Item Generation	Submission of the catalog item shall generate exactly one parent <code>sc_request</code> record and one child <code>sc_req_item</code> record. The parent request <code>price</code> shall reflect the aggregated cost of the selected laptop hardware bundle.	Must Have	Submit catalog order; query database to confirm generation of linked <code>sc_request</code> and <code>sc_req_item</code> ; verify record numbering conventions (<code>REQ...</code> , <code>RITM...</code>).
FR-04	Flow Designer Autonomous Triggering	The flow <code>Automated_Standard_Laptop_Fulfillment</code> shall trigger autonomously and execute within 5 seconds of <code>sc_req_item</code> insert when: <code>cat_item</code> IS "Standard Laptop Order" AND <code>state</code> IS "1" (Open).	Must Have	Inspect <code>sys_flow_context</code> upon record insertion; verify trigger conditions evaluate to true; verify execution begins within 5000 milliseconds.
FR-05	Dynamic Two-Tier Approval Routing	The flow shall execute the <code>Ask for Approval</code> action targeting the requester's direct manager (<code>requested_for.manager</code>). If <code>hardware_bundle</code> == <code>'Developer Standard (macOS)'</code> (unit cost > \$2,500), the flow shall dynamically append a secondary approval tier to the IT Finance Director.	Must Have	Submit Windows Developer order -> verify single manager approval generated. Submit macOS Developer order -> verify sequential or parallel approval generated for Finance Director.
FR-06	Actionable Interactive Notifications	The platform shall dispatch actionable email notifications to designated approvers containing employee details, hardware tier, cost center, and business justification, featuring 1-click cryptographic <code>[Approve]</code> and <code>[Reject]</code> actions requiring zero portal login.	Must Have	Trigger approval; inspect outbound email in <code>sys_email</code> ; reply via 1-click token; verify <code>sysapproval_approver</code> transitions state to <code>approve</code> automatically.
FR-07	Deterministic Rejection Handling	If any approval step evaluates to "Rejected", the flow shall immediately update <code>sc_req_item.state</code> = 4 (Closed Incomplete), set <code>stage</code> = <code>request_cancelled</code> , log the approver's mandatory comments in Work Notes, send an explanatory notification to the requester, and terminate execution.	Must Have	Reject approval record with reason "Budget exceeded"; verify RITM transitions to Closed Incomplete; verify requester receives rejection email; confirm zero <code>sc_tasks</code> records created.
FR-08	Automated Catalog Task (<code>sc_task</code>) Dispatch	Immediately upon approval confirmation, Flow Designer shall instantiate an <code>sc_task</code> assigned to the <code>Hardware Fulfillment Depot</code> group. The task must inherit all RITM variables into its Variable Editor and set Priority = 3 (Moderate) or 2 (High if requester is VIP).	Must Have	Approve pending RITM; verify <code>sc_task</code> generated within 1 seconds; verify assignment group is <code>Hardware Fulfillment Depot</code> ; verify short description includes laptop model and user name.

FR-09	Mandatory Asset Reconciliation (alm_hardware)	A ServiceNow Data Policy shall prevent closing an <code>sc_task</code> unless <code>u_asset_tag</code> is populated with a valid barcode corresponding to an in-stock asset in <code>alm_hardware</code> . Upon valid task closure, the flow shall update the asset record: <code>install_status = 1</code> (In Use), <code>assigned_to = requested_for</code> , <code>substatus = NULL</code> .	Must Have	Attempt task closure with error asset tag -> verify blocked with error. Enter valid asset tag <code>AST-009482</code> and close -> verify <code>alm_hardware</code> update status to <code>In Use</code> and sets <code>assigned_to</code> .
FR-10	End-to-End Portal Stage Transparency	The flow shall update <code>sc_req_item.stage</code> at each operational milestone: <code>waiting_for_approval</code> -> <code>fulfillment</code> -> <code>delivery</code> -> <code>complete</code> . These stages must render linearly on the Service Portal request tracker widget.	Should Have	Monitor Service Portal /esc during workflow execution; verify stage tracker reflects current phase accurately in real time.
FR-11	Cascading Record Lifecycle Closure	When the fulfillment <code>sc_task</code> transitions to <code>Closed Complete</code> , the flow shall update <code>sc_req_item.state</code> to <code>3</code> (<code>Closed Complete</code>) and trigger cascading closure of the parent <code>sc_request</code> if all child items are complete.	Must Have	Mark staging task Closed Complete; verify RITM transitions to Closed Complete; verify parent <code>sc_request</code> transitions <code>request_state</code> to <code>closed_complete</code> .
FR-12	Automated CSAT Feedback Dispatch	Exactly 2 hours following item closure, the system shall trigger an automated Customer Satisfaction (CSAT) survey containing 3 standard Likert-scale questions to the <code>requested_for</code> user.	Should Have	Fast-forward scheduled job / inspect <code>asmt_assessment_instance</code> verify survey record generated and invitation email sent to employee.

3. Non-Functional Requirements (NFR)

The system complies with 6 enterprise non-functional dimensions governing reliability, scale, compliance, and user experience.

3.1. NFR-01: Performance & Execution Latency

- **Trigger Initiation:** Flow Designer engine shall evaluate trigger conditions and instantiate `sys_flow_context` within **5 seconds** of `sc_req_item` database insertion.
- **Task Provisioning:** Automated creation of child `sc_task` records upon approval receipt shall complete within **10 seconds**.
- **Client-Side Form Performance:** The "Standard Laptop Order" catalog form shall reach Time-to-Interactive (TTI) within **1.5 seconds** over standard enterprise network connections (10 Mbps+), with client scripts completing execution within **200 milliseconds**.
- **Database Query Budget:** No catalog script or flow action may execute un-indexed table scans or queries taking longer than **100 milliseconds** of execution time.

3.2. NFR-02: Security, Governance & Role-Based Access Control (RBAC)

- **Principle of Least Privilege:**
- **Requesters** (`snc_internal`): Can view and track only requests where `opened_by == gs.getUserID()` or `requested_for == gs.getUserID()`.

- **Approvers** (*approver_user* or *itil*): Granted update access exclusively to approval records (*sysapproval_approver*) assigned to their user ID or delegated groups.
- **Hardware Technicians** (*itil*): Restricted to reading and updating *sc_task* records assigned to *Hardware Fulfillment Depot*.
- **Asset Synchronization**: Direct write operations to *alm_hardware* are restricted to the Flow Designer system execution context (*System User*) and credentialed Asset Managers (*asset* role).
- **Data Sanitization**: All user inputs in multi-line fields (*shipping_address*, *business_justification*) shall be sanitized against cross-site scripting (XSS) and SQL injection prior to database persistence.

3.3. NFR-03: High Availability, Fault Tolerance & Idempotency

- **Idempotent Flow Execution**: The flow design must be strictly idempotent. If a network interruption or node restart causes a flow action to re-fire, the system must query for existing child records before executing creation logic, preventing duplicate *sc_task* or approval generation.
- **Transient Error Handling**: Outbound notification actions shall incorporate automatic retry mechanisms (up to 3 retries at 60-second intervals) upon encountering transient SMTP or integration gateway failures.
- **Execution State Persistence**: Every state transition must be committed to *sys_flow_context*, allowing administrators to pause, resume, or replay flows without data loss.

3.4. NFR-04: Usability & Multi-Device Accessibility

- **WCAG 2.1 AA Compliance**: The catalog interface, variable controls, contrast ratios, and screen-reader labels must pass Web Content Accessibility Guidelines (WCAG) 2.1 Level AA standards.
- **Mobile Responsiveness**: 100% of ordering, approval, and task fulfillment views must render responsively within the native ServiceNow Now Mobile and Mobile Agent iOS and Android applications.
- **Zero Technical Jargon**: All catalog prompts and variable tooltips must use plain business language (e.g., "Developer High-Performance Laptop" instead of "Latitude 7440 Intel i9 13800H 32GB LPDDR5x").

3.5. NFR-05: Auditability & Compliance Logging

- **SOX / ITIL Audit Compliance**: 100% of approval submissions, rejections, delegation updates, and technician staging handoffs must be immutably preserved with microsecond timestamps in *sys_audit*, *sysapproval_approver*, and the RITM activity stream.
- **Zero Log Deletion**: Approval and asset attribution records are classified as permanent enterprise compliance artifacts and must be protected against manual deletion or truncation by Business Rules and Access Control Lists (ACLs).

3.6. NFR-06: Maintainability, Upgradeability & LTS Longevity

- **Zero Custom Code**: The implementation shall avoid custom compiled Java, jelly scripting, or deprecated Legacy Workflow Editor components.
- **Low-Code Architecture**: All process automation shall be constructed using native out-of-the-box Flow Designer action blocks, subflows, and standard ServiceNow data models.
- **Upgrade Safety**: Configurations must achieve a 100% pass rate in the ServiceNow Automated Test Framework (ATF) across Utah, Vancouver, Washington DC, and Xanadu LTS releases without requiring upgrade-skips or manual remediation.

4. Platform Technical Constraints & Design Assumptions

4.1. Technical Constraints

- ServiceNow LTS Platform:** The architecture is constrained to standard functionality available in ServiceNow Utah, Vancouver, Washington DC, and Xanadu releases.
- Core Plugin Dependencies:**
 - `com.glide.hub.flow_designer` (Flow Designer Engine v2.0+)
 - `com.snc.service_catalog.core` (Service Catalog Framework)
 - `com.snc.asset_management` (ITAM Hardware Asset Lifecycle Management)
- Execution Script Constraints:** Catalog Client Scripts must be configured as `Isolate Script = true` to run cleanly within the modern Service Portal Angular framework, avoiding direct DOM manipulation (`document.getElementById`).
- Data Policy Enforcement:** Validation of mandatory asset tags on `sc_task` must be enforced at the server-level via Data Policy (`sys_data_policy2`) to ensure API and REST updates cannot bypass the rule.

4.2. Design Assumptions

- Corporate user identity, manager hierarchies, and department assignments are synchronized continuously from enterprise Active Directory / Okta via automated LDAP/SSO sync into `sys_user`.
- Standard hardware bundles (models, RAM, CPU) are maintained in `cmdb_hardware_product_model` and kept in sufficient safety stock (minimum 15 units per model) at the central hardware depot.
- Technicians possess handheld barcode scanners capable of scanning Code 128 / QR asset tags directly into the browser or Now Mobile app.
- Line managers have active Microsoft Outlook / Exchange or mobile clients capable of rendering HTML5 Actionable Email Messages.

5. Service Level Agreement (SLA) & Operational Level Agreement (OLA) Hierarchy

To guarantee the primary business outcome of **reducing standard laptop procurement cycle time from 14.2 business days to under 3 business days (<72 hours)**, a multi-tier SLA/OLA hierarchy is established within the ServiceNow SLA Engine (`contract_sla` and `task_sla`).

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+
| SLA / OLA TIMELINE HIERARCHY (< 72 BUSINESS HOURS TOTAL)
|
+-----+
+
| [Overall SLA-01: End-to-End Procurement Request SLA (72h)]
|
| └─ [OLA-01: Line Manager Approval OLA (24h)]
|
|   └─ 0h: Approval Dispatched
|
|   └─ 24h: Automated Reminder
```



5.1. SLA / OLA Matrix & Governance Rules

Contract Code	Tier	Name	Target Duration	Schedule	Start Condition	Pause Condition	Stop Condition
SLA-01	Client SLA	End-to-End Laptop Procurement SLA	72 Business Hours	8x5 Corporate Business Schedule	sc_req_item created AND cat_item IS "Standard Laptop Order"	approval == 'requested' (Paused pending manager action)	state (Close Compl Close Incom
OLA-01	Internal OLA	Line Manager Approval Turnaround	24 Business Hours	8x5 Corporate Business Schedule	sysapproval_approver created AND state == 'requested'	None	state (appro rejec cance

FR-10	Stage Progression	US-08	sc_req_item.stage	NFR-04	Test Case TC-07
FR-11	Cascading Closure	US-08	sc_req_item, sc_request	SLA-01	Test Case TC-07
FR-12	CSAT Dispatch	US-08	asmt_assessment_instance	N/A	Test Case TC-08
NFR-01	Latency & Performance	All	Engine subsystem	SLA-01	Performance Benchmark PB-01
NFR-02	Security & RBAC	All	sys_security_acl	N/A	Security Audit SA-01
NFR-03	Fault Tolerance	All	sys_flow_context	SLA-01	Resiliency Test RT-01
NFR-04	WCAG & Mobile	US-01	Portal / Now Mobile	N/A	Accessibility Audit AA-01
NFR-05	Auditability & Logs	US-03, US-07	sys_audit, sysapproval_approver	N/A	Compliance Audit CA-01
NFR-06	Maintainability	All	Update Set Artifacts	N/A	ATF Test Suite ATF-01

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End of Solution Requirements Specification

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